OPERATION MANUAL FOR MICRO SCAN METAL DETECTOR

UNIQUE EQUIPMENTS

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You can help to improve this manual. If you find any mistakes or if you know of a way to improve the procedures, please let us know. Mail your comments to ***UNIQUE EQUIPMENTS***, Z-15, Marmugoa Indl. Area, Off. Airport Road, Chicalim, Goa 403 711.

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CHAPTER 1

GENERAL INFORMATION

1.1 PURPOSE

The Metal Detector system was developed with the objective of minimizing the threat to the modern equipments from damage due to the enter of foreign metals while in production. Modern industries that are concerned with improving the product quality need a device to monitor the product process to ensure that no stray metals contaminates the final product. Metal can enter the product process at any point. From incoming raw materials to the wearing and breaking of any part from the extension machinery's employed by today's modern manufacturers. This is were the Metal Detector come to be extremely employed by companies all over from the Food processing Industries to Pharmaceuticals and others Industries.

1-2 GENERAL OPERATIONS

It must be remembered all the times, however, that the metal detector can never be more than a tool! Design and manufacture of a quality instrument provide it with a unique ability and numerous performance capabilities. The effectiveness of any metal detector as a protection tool, however, is totally dependent upon the skill of the individual using it and the manner in which it is integrated with other systems.

No metal detector can ever be expected to function 100% efficiency when it is installed near high-powered electrical apparatus or next to a large and moving metal objects. Also detectors will usually not be as effective if operated immediately adjacent to computer or telephone switching equipments. This condition of physics… not of design… affects every metal detector, regardless of all other factors.

CHAPTER II

INTRODUCTION TO METAL DETECTOR.

2.1 INTRODUCTION:

Unique Metal Detector provide for the first time to the Indian industrialists world-class detectors, at reasonable cost. The state of art technology provides sensitivities for metal even much smaller than a pinhead. These Metal Detectors are built to last and will provide years of trouble free service in harsh environments. Systems of various sizes, shapes, and design are custom made as per the client's requirements. For installation in hygienic conditions stainless steel finish is provided wherever required. Metal detectors operate on the principle of Eddy Current induced imbalanced in a set of mutually balanced coils. Frequency of 10 to 200 KHz is used to generate an Electro magnetic field in the aperture area. The system automatically maintains a high degree of balance, which is disturbed by the metal passing through the aperture resulting in a signal, which is amplified to operate on audiovisual alarm and reject mechanism.

2.2 HOW THEY WORK: -

Unique Metal Detectors detect unwanted or stray metal in moving bulk material, sheet or web material, or package or bagged material. They can also be used to detect the presence of metal item, which is intended to be in a non-metallic package (i.e. metal spout in a card board box or a metal premium in a cereal or food package.)

The Unique metal detector is installed around a conveyor or chute so that material or packages to be inspected will pass through the detector's aperture. The detector creates a high frequency Electro magnetic field through which all conveyed material and packages must pass. Presence of foreign metallic particles causes a reaction in this field. The detector senses this reaction and the signal is amplified as required. The signal output is in the form of a relay contact closure with two normally open and two normally closed contacts. The output may be used to stop a conveyor, sound an alarm, and actuate a marking system or any other device or combination of devices. The operating mode indicate that the conveyor is to be stopped upon detection of metal, a spray marking system or a trip counter should be used to indicate the number of pieces of metal passing through the detector while the belt is stopping. Most Unique metal detectors can be used with conveyor speed from 0.1 Mts./sec. onwards without any special modifications. All models are designed to provide the greatest possible sensitivity while maintaining stable operation.

2.3 PRINCIPLE OF OPERATION:-

The search coil consists of three equally spaced coils. The center winding is connected to an oscillator and so sets up an electromagnetic field, which induces voltages into the either windings. These outer windings are connected in opposition & so the induced voltages are balanced to zero. The products to be inspected are passed along the common axis of these coils. The introduction of a piece of metal in to the aperture causes the induced voltages to be unequal. The resultant out of balance signal is amplified in the control unit and the relay is operated, so operating any rejection or alarm system, which may be connected to it.

2.4 SAILENT FEATURES

2.4.1 Automatic Balance Principle

2.4.2 Improved and compact Search Coil

2.4.3 Membrane Touch Pad With 04 Keys With Tamper Proof Controls & Password protection against unauthorized use.

2.4.4 Continuous Self-check and calibration verification.

2.4.5 A LED Bar Graph Provided to display continuously Status of Coils.

2.4.6 100 Product memories.

2.4.7 Integral alarm, On Delay and Off Delay timings.

2.4.8 20 Characters X 4 Lines LCD Display for clear information display.

2.4.9 All sensitivity check, settings and routine QC checks are recorded.

2.4.10 Solid-state relays for arc free switching.

2.4.11 Internal data store & Clock.

2.4.12 Quick recovery after the detection of tramp metal.

2.4.13 Automatic and manual reject rest facility.

2.4.14 Reject Confirmation (Optional)

2.4.15 Detection and rejection counters. (Optional)

2.4.16 Logging Capability to a Printer for HACCP and ISO 9001 Documentation

2.4.17 Parallel Printer / RS 232 Printer Output. (Optional)

CHAPTER III

SYSTEM COMPONENTS

3.1 SEARCH HEAD

The Search Head is a part of metal detector, which creates the required electro magnetic field and senses disturbances caused by the presence of metal in the material passed through its aperture.

3.1.1 It is an all-metal construction except for the aperture section. It is made as rigid as possible to limit influences of shocks and vibrations.

3.1.2 The aperture area is of a bonded material and has coils around it for the required Electromagnetic field based metal detector.

3.2. CONTROL SECTION

3.2.1 It generates all the power supplies required and processes the signal from the Search Head to annunciate audiovisual indications and other control Systems. This is placed inside the mounting base panel of the Search Head. Proper Earthing should be given to the Metal Detector to function.

CHAPTER IV

4.1 SYSTEM CONTROLS INDICATIONS:

The following controls, indications and terminals are provided in the metal detector system.

4.1.1 **POWER ON**:

This is to indicate the application of mains power to the unit. This indication is installed at the base of the metal detector-mounting box.

4.1.2 **METAL DETECT**:

This is a LED based visual indication which indicate when a metal is passed through the metal detector aperture. This indication delay can be controlled by the ON Delay Timer which is provided in the timer section of the metal detector.

4.1.3 **METAL REJECT**:

This is a LED based visual indication which indicate the rejection of the detected metal Contamination by the metal detector aperture. This indication delay can be controlled by the OFF Delay Timer which is provided in the timer section of the metal detector.

4.1.4 **REJECTION ERROR**:

This is a LED based visual indication, which indicate the status of the metal detector. This indication glows when the Detect and Reject count doesn't mach each other and gets off when the fault is removed or acknowledged.

4.1.5 **BYPASS** :

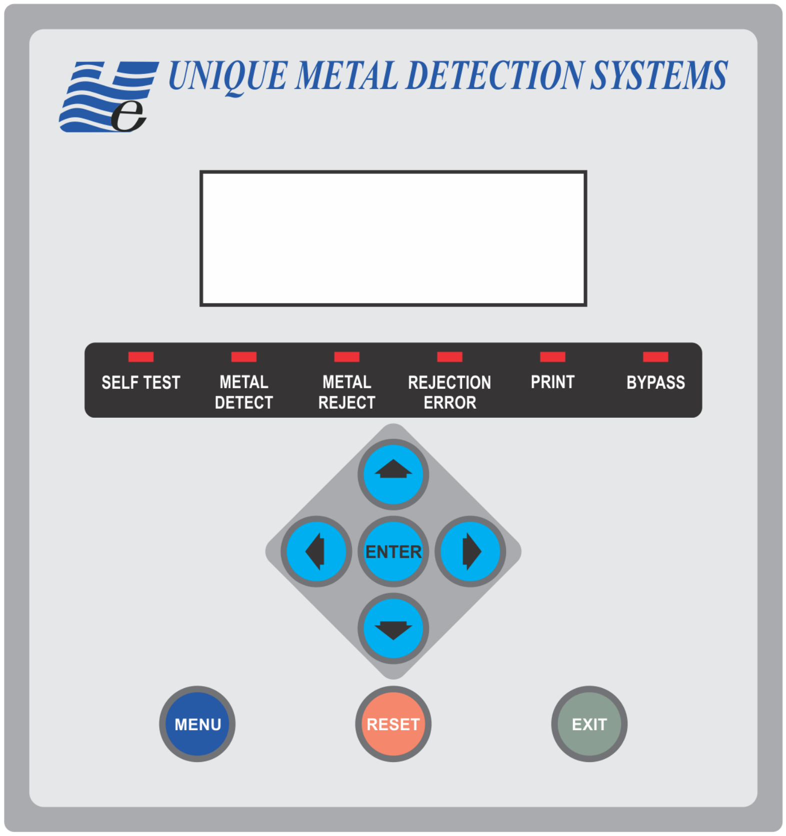
This is a LED based visual indication, which indicate the status of the metal detector. This indication glows when the Bypass Switch is operated which is installed inside the control box of the meal detector. This indication gets off when the Bypass switch is made Off.

4.1.6 **PRINT** :

This is a LED based visual indication, which indicate the status of the printer. This Indication will glow till the entire printing job is carried out.

4.1.7 **SELF TEST** : This is a LED based visual indication, which will be ON once the detector tests of its own for any irregularities. This will glow according to the time settled duration. The regular practice is once in Every Hour.

4.2 **CONTROL KEYS**:



4.2.1 **ENTER**:

To enter a selected menu/sub menu/value.

4.2.2 **UP KEY:**

This key is to increase (Increment) the reading, which is displayed in the LCD display unit. E.g. To increase the Sensitivity, timer, phase e.t.c.

4.2.3 **DOWN KEY**:

This key is to decrease (Decrement) the reading, which is displayed in the LCD display unit. Eg. To increase the Sensitivity, timer, phase e.t.c.

4.2.4 **LEFT KEY & RIGHT KEY**: To move the cursor left or right.

4.2.5 **MENU**:

This key is used to enter into Main Menu (to be pressed for 4 secs.)

4.2.6 **RESET**:

Used to enter supervisor setting directly.

4.2.7 **EXIT**:

Used to exit Main Menu / Sub Menus.

HOW TO SWITCH *ON* THE METAL DETECTOR?

The mains ON / OFF switch is provided at the mounting base of the metal detector. Press the switch to On the Metal Detector.

4.4 **HOW TO SET THE METAL DETECTOR?**

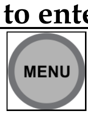
Once the system is switched On the display provided in the front panel of the metal detector shows some readings and settings. They are listed below.

UNIQUE METAL
DETECTION SYSTEMS
Initializing…05

Wait for 05 Seconds, after 05 Seconds the display shows as below…

PRODUCT CODE	: 000
SENSITIVITY	: 100
COUNT	:0000/0000

How to enter Main Menu:

Press 

key for 4 Sec to enter **MAIN MENU**.

→ OPERATOR SETTING←
SUPERVISOR SETTING
RESET COUNTER
PRINT

Press  to move the cursor to desired location on the LCD [For instance SUPERVISOR SETTING].

OPERATOR SETTING
→ SUPERVISOR SETTING←
RESET COUNTER
PRINT

<u>How to enter SUPERVISOR SETTING:</u>

OPERATOR SETTING
→ SUPERVISOR SETTING←
RESET COUNTER
PRINT

Press ENTER to enter SUPERVISOR **SETTING**.

SUPERVISOR PASS

0000

Pressing ENTER, and combination of ▲. ▼. ◀. ▶. After entering set password press ENTER to enter the set value into the system. Password by default is 0000. The system will enter/display a Sub Menu as below.

→ EDIT PRODUCT ←
EDIT OPERATOR
SELF TEST
SELF TEST TIMER

Above Menu is only accessible to a Supervisor.

Press ENTER to Edit Product. Press ▲. ▼ to choose desired Sub Menu. Choosing **SELF TEST**, machine will perform a self test, **SELF TEST TIMER** can be set from 001 to 255 min.

<u>How to Edit Product:</u>

Selecting **EDIT PRODUCT** will take you to a Sub Menu to Edit Product. The Menu is as follows.
ENTER PRODUCT CODE (Default is 000) desired product code can be entered using . And pressing .

This will take you to a following menu.

PRODUCT NAME
SENSITIVITY
PHASE
FLAP OPEN TIME
POST DETECT DELAY
PRE-REJECT DELAY

By pressing the above sequence is displayed. If is pressed the above sequence will be displayed from bottom to top.

The values can be edited (Individually) by pressing , and combination of . . . . After choosing desired value press to enter the set value into the system.

At any point pressing will get you back to the previous menu. Supervisor can also edit Operators (Can edit Operator Code and Passwords).

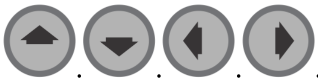
Choosing **EDIT OPERATOR** will take you to a Sub Menu, **OPERATOR CODE**. The value can be edited (Individually) by pressing , and combination of . . . .

After choosing desired value press to enter the set value into the system.

This will take you to a Sub Menu as follows:

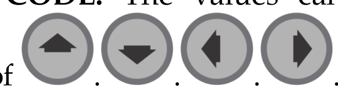
OPERATOR NAME

OPERAROR PASSWORD

The values can be edited (Individually) by pressing ENTER, and combination of . After choosing desired value press ENTER to enter the set value into the system.

<u>How to enter OPERATOR SETTINGS:</u>

```
→ OPERATOR SETTING←
  SUPERVISOR SETTING
    RESET COUNTER
        PRINT
```

Choose **OPERATOR SETTING** by pressing ENTER. This will take you to a Sub Menu **OPERATOR CODE.** The values can be edited (Individually) by pressing ENTER, and combination of . After choosing desired value press ENTER to enter the set value into the system. After this the will ask for **OPERATOR PASSWORD**(by default it is 0000). The operator can choose only **PRODUCT CODE** using above mentioned sequence of keys. After this the operator can view the following Sub Menu.

PRODUCT NAME
SENSITIVITY
PHASE
FLAP OPEN TIME
POST DETECT DELAY
PRE-REJECT DELAY

Now the metal detector is ready for its operation. The user can set the metal detector for 100 Products (000 – 099). We recommend to pass the test samples through the metal detector after each and every product set up and all the test samples should be detected and rejected. When the machine is set and/or in operation following will be displayed on the screen.

PRODUCT CODE : 000
SENSITIVITY : 100
COUNT :0000/0000
▮▮

<u>How to RESET COUNTER:</u>

OPERATOR SETTING
SUPERVISOR SETTING
→ RESET COUNTER←
PRINT

Press 

for 4 secs. To enter main Menu. Use  to select RESET COUNTER, press 

. Confirmation is asked by the system , press 

to confirm. Counters are reset to 0000 /0000.

OPERATOR SETTING
SUPERVISOR SETTING
→ RESET COUNTER ←
PRINT

<u>Note: If rejection error LED glows press  for 4 secs. LED will switch off.</u>

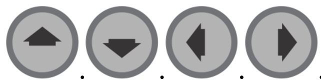
How to take Printout.

OPERATOR SETTING
SUPERVISOR SETTING
RESET COUNTER
→ PRINT ←

Press  for 4 secs. To enter main Menu. Use  to select **PRINT**, press  this will take you to following option:

ENTER RECORD DATE

09/09/10 - 09/09/10

pressing , and combination of . After choosing desired date press .

It will show the following on the display

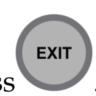
PRESS SELECT TO CONFIRM
ACTION

press . Confirming this the system will start printing and display the following

Printing ……

After the system stops printing (end of Record Date) the system will display Main Menu

OPERATOR SETTING
SUPERVISOR SETTING
RESET COUNTER
→ PRINT ←

To come out of this Menu please press . The system will be in operation mode.

Please note that Unique Metal Detector supports **EPSON – LX 300**

CHAPTER V

INSTALLATION: -

5.1 Mounting.

The Search head should be positioned so that wherever possible the product passes through the head evenly distributed around the center of the aperture. Precautions must be taken to ensure that the head is not subjected to sudden movements. Such movements do not usually occur if the support is fairly rigid. They may, however result from the operation of heavy reject mechanism or from associated machines. Anti vibration mounts usually produce little improvement. A better approach is to build a separate sub-frame for the head. This is only required in very exceptional circumstance.

5.2 Metal Free Area for Metallic items moving near the Search Head.

It is difficult to predict the minimum distance between the search head and a moving metal.

This will depend on the following parameters:-

a) The Search Head Aperture.
b) Sensitivity setting
c) The size, shape and composition of the moving metal.
d) The position, direction and speed of the movement of the moving metal

5.3 General precautions

The belt should be plastic coated (i.e. Non metallic) and should be free from all metallic joints and inclusions. All rollers should have a non-metallic surface otherwise metal will gradually be transferred to the belt.

5.4 Clearances in the Metal Detector Aperture

The conveyor belt, skid plate, product guides or the product itself must be completely free of all metal and must not touch the metal detector aperture in any way whether on the top, bottom or sides of the aperture. Care should be taken to ensure that the skid plate does not distort under load and touch the bottom of the aperture. A buildup of product under the skid plate is a typical cause of false triggering. The belt must not be allowed to wander from side to side and rub on the inside of the aperture as this could damage the aperture lining. Before fitting a skid plate or product guide it is advisable to check that these are completely metal free by passing them through the Metal Detector. Suitable skid plate and product guide materials include Polyethylene, Polypropylene, P.V.C., Wood and wood laminates.

5.5 Electrical Loop

This is one of the largest single causes of faulty metal detector operation. The search coil in a metal detector generates a high frequency in the aperture, the metal case of the detector acts as a screen to prevent metal outside the detector head affecting the search coil. Some of the high frequency electrical fields from the detector head however escape from the aperture through which the product passes. This can cause very small electrical current to flow in nearby metal structures, which form closed electrical circuits and take power from the search coil system. This cause no problem if the loops are completely closed or completely open but if the electrical path is intermittent then false triggering of the detector is likely. Typical causes of intermittent electrical loops include loose bolts on the conveyor or on the metal detector mounting brackets, corrosion of metalwork, broken welds, open hinged doors, conveyor idler rollers, broken or rubbing contacts. Interference can be overcome by opening the conductive path by using an insulated pad or closing the path by welding or tightening the bolts so that it cannot become intermittent. Loop problems in rollers can usually be overcome by mounting the idlers rollers closest to the detector on an insulating block. The metal bolts joining the roller to the insulating block should not make contact with the conveyor or detector head. An intermittent loop on a conveyor may also due to lubricated bearings whose balls act as electrical contacts and therefore whose resistance varies as they move through the lubricant. The source of such loop interference can be very elusive and difficult to locate. The higher the aperture the greater the high frequency leakage our of the aperture and the greater the possibility of trouble from loops.

5.6 Metal objects near the Metal Detector

The metal detector is very efficiently screened and large masses of metal near the top, bottom and ends will not effect the detectors performance. However metal positioned close to the aperture can cause interference problems if it moves or vibrates. The area close to the aperture, which should be kept metal free, is known as Metal Free Zone. The metal free zone is dependent upon the aperture dimensions and the sensitivity setting of the detector.

Metal, or metal rollers close to the aperture act as aerials and pass small electrical currents into the conveyor framework, this result in some or all of the following effects:
The detector head becomes more sensitive to vibration. If the detector head is accidentally bumped it will cause relative movement between the detector and the offending metal so triggering the detector.

The whole framework may become more sensitive to vibration. If the detector is able to twist or flex or if there are some loose or badly welded joints the electrical currents will be intermittent so triggering the detector.

5.7. Electrical Interference:-

It is always good practice to suppress electrical interference at its source and if trouble is experienced the offending source should be located and suppressed. If this is not possible, it may be necessary to operate the detector at reduced detection sensitivity. The relay in the metal detector is often used to control solenoid valves, power contactors and similar electrical devices, when these are switched off the collapsing field of the inductive winding generates a wide band radial frequency interference which may be picked up by the search coil system in the detector head and cause false triggering. A capacitor of suitable capacity and working voltage connected across the offending device to ground as close to the device as possible will often provide satisfactory suppression.

It is very important to keep electrical interference away from the search coil in order to ensure adequate sensitivity and false alarms. The most important measures to prevent electrical interference are:-

Much can be learned, if the unit can be checked out while the conveyor on which it is mounted and other equipments in the adjacent area is shut down. If the metal detector operates properly under this condition, it would verify that the false signals are external. However if poor operations continuous the problem may be power line disturbance.
Some common source of external interference are moving or vibrating metal objects too close to the search coil.

Metal particle embedded in the conveyor belt, or metal fasteners should be located by running conveyor empty and observing it for a complete cycle of its total length through the Metal Detector, this will facilitate in searching for any tramp metal if present in the belt which will get detected when passed through the search coil.

Excessive vibration of the search coil and control unit.

Metal cover plates, metal roofing and metal guide plates between the adjacent roller stages must be removed or replaced by non-metallic parts.

Deck plates should be welded to the frame if they extended in this area.

The adjacent roller stages are to be welded to both their support beams, if they are already bolted on their support, short welding joint will be sufficient (about 15mm)

All bolted & riveted connections with in 2mtrs from the search coil should be secure with short welding joints.

Note: Bolted and riveted connections are unpredictable contacts for electrical currents that are known to raise problems in the vicinity of the search coil. The additional short welding joints will relieve these problems.

CHAPTER VI

6.1 OPERATING INSTRUCTIONS

6.1.1 Allow a period of up to about 5 minutes.

6.1.2 In stabilised state, there will be NULL indication on the Meter / Bar Graph. Allow a test piece to pass through the aperture of the metal detector and adjust the sensitivity control such that its gets adequately detected.

With the help of Timer Control Set the time the metal detector should remain tripped after detection of metal. On lapse of the time the trip lamp would go out.

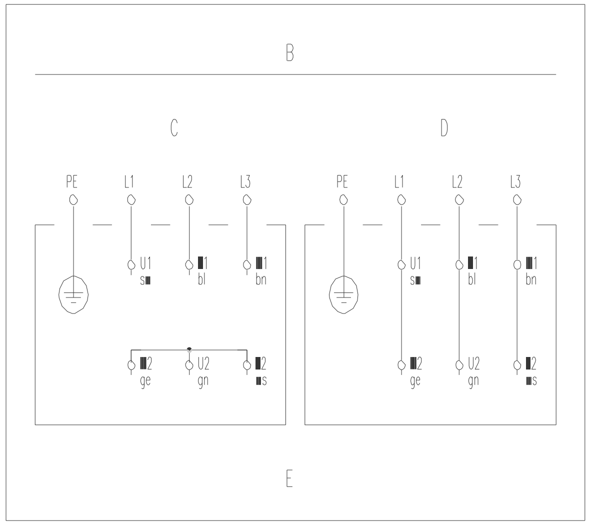
Sometimes the product itself has a capacity to generate signal of its own which is normally due to conductive materials, the product is sometimes made up of, this effect which can cause false triggering of the unit can be reduced by adjusting the PHASE control. The adjustment can be made such that the ratio of the signal due to product with metal and signal due to product without metal contamination is reasonably large. The adjustments as regards to sensitivity and phase controls are generally not required to be re-done expect when there is an increase in level vibrations from the environment. In the later case, of course, dealing with the vibrations will be more of a preferred approach.

STANDARD OPERATION PROCEDURE FOR INSTALLING THE METAL DETECTOR.

1. Identify the place where the Metal Detector have to be installed and place the same accordingly.
2. Isolate the Metal Detector frame from the conveyor frame. (By removing the metallic strips, marked in yellow color)
3. Check the leveling pad, adjust it properly.
4. Check all the screw, nuts and all other accessories are properly connected.
5. Connect the single phase to Metal Detector and Three Phase to Conveyor Starter.
6. Switch on Conveyor belt
7. Check for the conveyor alignment. If not aligned , align it properly by tightening the alignment studs fitted at both the sides of the conveyor.
8. After starting the starter kindly check the direction of the belt as indicated by the Arrow.
9. Switch ON the system.
10. Wait for about 25 – 30 sec till the (Siren Stops). Do not pass any product while siren is buzzing.
11. Pass the test samples, it should be detected.
12. Pass the product from the in feed of the conveyor.
13. If metal is detected inside the product the conveyor belt will stop indicated by siren lamp remove the bag and keep it on the rejected area.

Connecting plans motors

Three phase motor:



B Connection to the 3-conductor-mains
C Star connection (high voltage)
D Triangle connection (low voltage)
E For switching over the direction of rotation two phases of the supply must be exchanged!

2. Safety instructions

The conception and production of our conveyor belts has been carried out very carefully, in order to guarantee trouble-free and safe operation. Please read the operating instructions completely, before starting the machine. Always observe the safety instructions.

Make sure that all persons working with or at this machine have carefully read the following safety instructions.

When using the conveyor belt in humid or wet environment (wet area) it has to be made sure that the required insulation type is provided.

Starting, modification, maintenance and repair work may only be carried out by skilled and authorized personnel.

Disconnect the mains power supply during assembly, maintenance and repair.

Work at the electric equipment may only be carried out by an electrician or a person trained in electrical engineering under the supervision of an electrician, according to the electronics regulations.

- User and operator must take care that only authorized personnel works at the conveyor belt.
- The operator must immediately be informed about any changes impairing safety.
- Modifications impairing safety must immediately be reported to the operator.
- The conveyor belt may only be operated according to the intended use.

Dangers occurring at the machine

- In case the conveyor belt gets wet, there is the danger of an electric shock!
- Make sure that the protector ground of the electric power supply is in perfect condition!
- Operation of the conveyor belt without trim panels and protection hoods (chain drive) is prohibited in any case

Intended use

The intended use of conveyor belts is the transport of material.

The material to be transported must have a side length of at least 5 mm. By special designs or other equipment the conveyor belts can be retrofitted for material with smaller side length (>0.5 mm).
If necessary, please turn to the manufacturer

Mounting on stands

When mounting the conveyor belt on a supporting table, it has to be taken care that the feet of the stand are tightly screwed down to the table.
When mounting the conveyor belt on a stand, the feet of the stand must be additionally anchored in the foundation by means of dowls.

Switching on and off of the conveyor belt takes place at the protective motor switch

Adjusting the belt run

Motor and conveyor belt were tested at the manufacturer's plant and went through a final inspection. An adjustment of the belt run may become necessary when the conveyor belt is initially erected or owing to the running-in behavior of the belt.

Maintenance

When starting, maintaining and repairing the conveyor belt it must be all-polo separated from the mains. Work at electric equipment of the conveyor belt may only be carried out by an electrician or
by persons trained in electrical engineering under the supervision of an electrician according to the electronics regulations.

Belt

In case the belt is dirty, clean it with spirit and a non- fluffy cloth. For conveyor belts used for foodstuff use a permissible spirit substitute.

Motor

Gear motors are maintenance-free for 10.000 operating hours.

Depending on the dust deposit, clean the protective fan hood of the motor, the motor itself and the gear unit so that sufficient cooling of the drive unit is al- ways guaranteed.

Gear unit

When being delivered the gear units are ready for operation and are filled with gear lubricant and oil.

Consequently a long-term lubrication is guaranteed for all movable parts.

Dismounting, cleaning and changing the oil is not necessary.

Deflection, drive and supporting rollers

In case the rollers are dirty, clean them with spirit and a clean, non-fluffy cloth. For conveyor belts used for foodstuff use a permissible substitute.

Environmental influences

When erecting the conveyor belt, take care that the belts are not exposed to severe heat radiation. Observe the permissible temperatures of the belts (see brochure). Otherwise the belts can extend and slip through at the drive rollers.

Keep oil, chips, etc away from the conveyor belt.

Stock keeping of spare parts and after sales service

In order to guarantee quick and faultless handling of the order, please always state the type of equipment (see type plate), number of pieces needed, spare part name and spare part number.

VOLTAGE CHART FOR PCB MSMD – 001

PIN NOS.	SET CONDITION	TRIP CONDITION
1	11.42 VDC	11.42 VDC
2	NC	NC
3	NC	NC
4	NC	NC
5	GND	GND
6	01 V	01 V
7	10 V	10 V
8	12 V	0 V WITH METAL
9	02.23	2.23 PHASE
10	02.23	2.23 PHASE
11	0 mV	+/- 5 V SIGNAL
12	NC	NC
13	NC	NC
14	NC	NC
15	NC	NC
16	NC	NC
17	5.55 V	5.55 V REF VTG
18	-5.55 V	-5.55 V REF VTG
19	-11.91 VDC	-11.91 VDC
20	0 mV	0 mV
21	0 mV	0 mv
22	0 mV	0 mV

VOLTAGE OUTPUTS FOR IC'S OF PCB- MSMD-001.

PIN NOS OF LM 324	SET CONDITION		TRIP CONDITION	
	IC1	IC6	IC1	IC6
1	0 mV	0.2V	0 mV	0.2 V
2	0 mV	0 mV	0 mV	0 mV
3	0 mV	0 mV	0 mV	0 mV
4	11.82 V	11.82V	11.82 V	11.82 V
5	0 mV	0 mV	0 mV	0 mV
6	0 mV	0 mV	-0.76 V	0 mV
7	10.60 V	-1.26V	-10.60 V	-1.25V
8	0 mV	-1.5 V	0 mV	-1.5V
9	0 mV	0 mV	0 mV	0 mV
10	0 mV	0 mV	0.77 V	0 mV
11	-11.92V	-11.92V	-11.92 V	-11.92 V
12	0 mV	0 mV	-11.59 V	0 mV
13	0 mV	0 mV	-11.58 V	0 mV
14	0 mV	0 mV	-11.15 V	0 mV

VOLTAGE CHART FOR IC's FOR PCB MSMD-001.

	SET CONDITION				TRIP CONDITION			
Pin No.	IC2 4538	IC7 339	IC8 4031	IC9 555	IC2 4538	IC7 339	IC8 4031	IC9 555
1	0 mV	1.37V	-11.92V	-11.92V	-11.92V	1.37V	-11.92V	-11.92V
2	0 mV	0 mV	-5.2V	-5.84V	0mV	0mV	-5.20V	-5.92V
3	0 mV	11.81V	0mV	-5.25V	0mV	0mV	0mV	-5.92V
4	0 mV	0mV	0mV	0mV	-11.92V	0mV	0mV	0mV
5	0 mV	0mV	-5.2V	-3.94V	0mV	0mV	-5.22V	0mV
6	0mV	0mV	-11.91V	-5.84V	-11.92V	0mV	-11.92V	0mV
7	0mV	0mV	0mV	-5.83V	0mV	0mV	0mV	0mV
8	0mV	0mV	-11.91V	0mV	-11.92V	0mV	0mV	0mV
9	0mV	0mV	-5.23V		-11.89V	0mV	-5.23V	
10	-11.92V	0mV	-11.91V		0mV	0mV	-11.91V	
11	-11. 92V	0mV	0mV		-11.02V	0mV	0mV	
12	-5.23V	0mV	-5.9V		-5.23V	0mV	-5.23V	
13	0mV	1.45V	0mV		0mV	1.45V	0mV	
14	0mV	1.42V	0mV		-10.68V	1.42V	0mV	
15	-11.91V		-11.91V		-11.92V		-11.92V	
16	0mV		0mV		0mV		0mV	

VOLTAGE OUTPUTS FOR TRANSISTORS OF PCB- MSMD-001.

	SET CONDITION			TRIP CONDITION		
	E	B	C	E	B	C
Q1	0mV	0mV	-16.44V	0mV	-0.7V	0.13V
Q2	-15.38V	0mV	0mV	0mV	-0.68V	0mV
Q3	-15.46V	0mV	0mV	0mV	-0.68V	-7.58V
Q4	0mV	0.6V	3.73V	0mV	0.6V	3.71V
Q5	0mV	0.35V	0mV	0mV	0.35V	0mV
Q6	0.10V	0.70V	1.76V	0.10V	0.70V	1.76V
Q7	0mV	0.36V	0mV	0mV	0.36V	0mV
Q8	0mV	0.35V	0mV	0mV	0.35V	0mV
Q11	1.28V	1.01V	10.51V	1.28V	1.01V	10.51V

VOLTAGE OUT PUTS FOR FIELD EFFECT TRANSISTIORS IN PCB: MSMD-001.

FET.	S	D	G	S	S	D	G	S
Q9	0mV	0mV	0mV	0mV	0mV	0mV	0mV	0mV
Q10	0mV	0mV	-1.24	0mV	0mV	0mv	-1.24V	0mV
Q12	2.23v	9.58V	0mV	0mV	2.22V	9.58V	0mV	0mV
Q13	3.70V	11.83V	2.23V	0mV	3.69V	11.83V	2.23V	0mV

NOTE :- ALL VOLTAGES ARE WITH RESPECT TO GROUND.

6.3 TROUBLE SHOOTING OF METAL DETECTOR.

6.3.1 System does not function at all.

a) Check the main supply.

b) Check the fuse at the SMPS which is installed at the base of the metal detector mounting.

c) Check for loose connection.

d) Check the DC Voltage on PCB No. MSMD-001 Pin No.01 & 19 this should show +12VDc at Pin No.01 and –12VDc at Pin No.19 respectively.

6.3.2 If system is out of NULL.

a) Check +12V and –12V supply on PCB No. MSMD – 001 Pin No. 1 & 19 resp. if found faulty. Replace SMPS board.

b) Check the oscillation of Q11with the help of an Oscilloscope (CRO). Connect the probe to the collector of the Q11. The CRO reads sine wave oscillator amplitude of aprox. 25 to 50 Volts (Peak to Peak) if the waveform is absent replace the Transistor.

6.3.3 Metal Detector does not senses / detect metal pieces.

a) Check the sensitivity control level.

b) Check for FAULT indication at the display panel of the metal detector. If fault persist try pressing RUN and select the Product code. Also check for the DETECT COUNT & REJECT COUNT the both should be equal. If not set it equal by Pressing RUN Key

c) Check the voltages at R44 & R66 on PCB No. MSMD – 001. It should be in between 0.5Vdc to –2.5Vdc. If found out of range replace Q9/Q10 or IC6.

6.3.4 Metal Detector sense but does not reject

a) Check whether the solenoid is connected to the system. Check for the continuity of the solenoid.

b) Check transistor Q1 on PCB No. MSMD-001. If found faulty replace the same.

c) Check the voltages at terminal No. 1 & 2 of the 04 Pin connector which is installed at the base plate. It should be 200VDc when metal is detected.

d) If 200VDc is absent replace SSR (Solid State Relay) for Solenoid.

e) If 200V Dc is present and does not reject when metal is detected, then check the continuity for solenoid.

6.3.5 Metal Detector is in trip/detect mode.

a) Check the voltage of IC6 at Pin No. 14. It should be in zero volts. If out of range replace the same.

6.4 Maintenance and Cleaning of the Metal Detector:-

6.4.1 Unplug the equipment from mains supply before cleaning so that there is no possibility of an electrical shock.

6.4.2 Clean the equipment with the blower/ vacuum cleaner to remove the powder.

6.4.4 Remove the Teflon inlet chute and wipe it with soft damp cloth.

6.4.5 Clean the aperture with soft damp cloth.

6.4.6 Disconnect terminals of solenoid valve and dismantle the reject mechanism and wipe it's surface with the soft damp cloth.

6.4.7 Never use strong solvent.

6.4.8 Above procedure should be followed after each batch/ change of product or once in a week.

6.4.8 Ensure that excessive vibrations do not transmit to the Metal Detector stand.

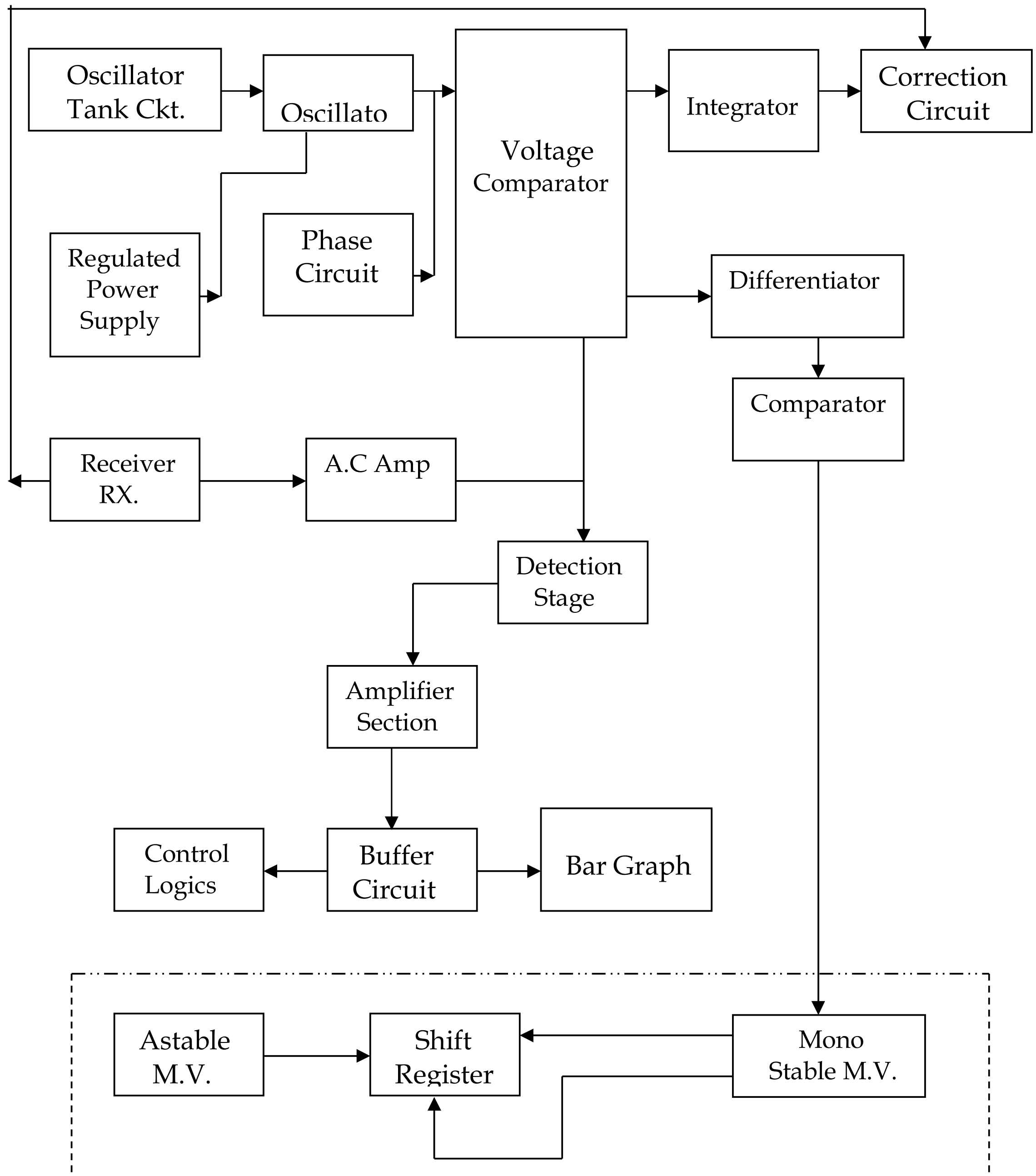
6.4.9 All the covers/panels provided on the Metal Detector should be kept closed after the cleaning.

6.4.10 Check the metal detector by passing the test piece in regular interval.

6.4.11 An electronic failure does not mean that the user has to run unprotected whilst trying to fix the detector or waiting for a technician. The exchangeable module gets the unit back into operation fast.

6.5: RECOMMENDED SPARES FOR TROUBLE FREE OPERATION OF THE METAL DETECTOR

DESCRIPTION	QTY.
6.5.1. PCB MOUDULE MSMD-001	1 No.
6.4.2. SMPS	1 No.
6.4.3. SSR	1 No.
6.4.4. MAINS FUSE LINK 100mA 30 mm.	1 No.
6.4.5 FUSE LINK FOR SOLENOID 1Amp 30mm	1 No
6.4.6 MAINS ROCKER SWITCH	1 No
6.4.7 MICRO CONTROLLER KIT	1 No
6.4.8 SOLENOID	1 No



BLOCK DIAGRAM